For instance, if we have two functions, say $f(x)=\sqrt{x+1},\ g(x)=2x-7$ and suppose that we want to evaluate these functions at say $x=3.25$. Which one will be easy to evaluate? Obviously, $g(3.25)$ will be easier to calculate than $f(3.25)$. If we are ready to accept some error in calculating $f(3.25)$, then we can find a linear function that approximates $f$ near $x=3$ and use this linear function to obtain an approximate value of $f(3.25)$.We know that the graph of a function is a nonvertical line if and only if it is a linear function. Out of infinitely many straight lines passing through any given point on the graph of the function, only tangent line gives a good approximation to the function, because the graph of $f$ looks approximately a straight line on the vicinity of the point (3,2).

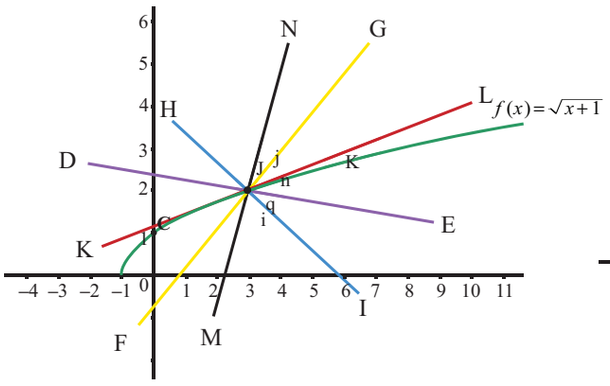



Fig. 8.1

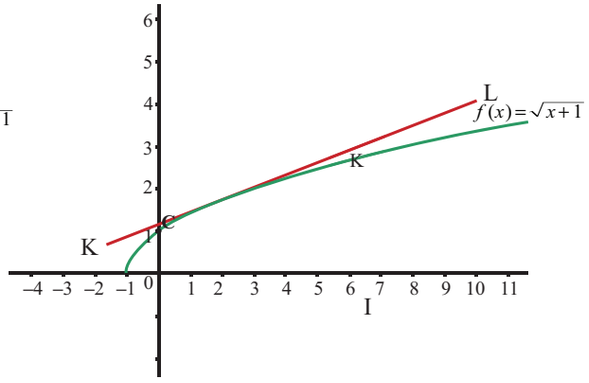



Fig. 8.2 Tangent Line

From the figures above it is clear that among these straight lines, only the tangent line to the graph of $f(x)$ at $x=3$ gives a good approximation near the point $x=3$. Basically we are **"linearizing"** the given function at a selected point $(3,2)$. This idea helps us in estimating the change in the function value near the chosen point through the change in the input. We shall use **"derivative"** to introduce the concept of **"differential"** which approximates the change in the function and will also be useful in calculating approximate values of a function near a chosen point. The derivative measures the instantaneous rate of change where as the differential approximates the change in the function values. Also, differentials are useful later in solving differential equations and evaluating definite integrals by the substitution method.

After learning differentials, we will focus on real valued functions of several variables. For functions of several variables, we shall introduce **"partial derivatives"**, a generalization of the concept of **"derivative"** of real-valued function of one variable. Why should we consider functions of more than one variable? Let us consider a simple situation that will explain the need. Suppose that a company is producing say pens and notebooks. This company is interested in maximizing its profit; then it has to find out the production level that will give maximum profit. To determine this, it has to analyze its revenue, cost, and profit functions, which are, in this case, functions of two variables (pen, notebook). Similarly, if we want to consider the volume of a box, then it will be a function of three variables namely length, width, and height. Also, the economy of a country depends on so many sectors and hence it depends on many variables. Thus it is necessary and important to consider functions involving more than one variable and develop the "concept of derivative" for functions of more than one variable. We shall also develop the concept of **"differential"** for functions of two and three variables and consider some of its applications. In this chapter, we shall consider only real-valued functions.



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